EXP: 7 ENTITY RELATIONSHIP (ER) DIAGRAM

AIM: To design and develop an ENTITY RELATIONSHIP (ER) DIAGRAM for the system "PRECISION DISASTER RECOVERY (PDR-SYNC)" representing entities, attributes, and relationships.

THEORY:

An ENTITY RELATIONSHIP (ER) DIAGRAM is a data modeling technique used to represent the structure of a database. It defines how entities are related to each other and what attributes they contain.

It is mainly used in:

- Database design
- System analysis
- Data modeling

An ER diagram helps in identifying:

- Entities (objects)
- Attributes (properties)
- Relationships (associations between entities)

COMPONENTS OF ER DIAGRAM:

ENTITY: Represents a real-world object or concept
Example: Responder, Report, Verification

ATTRIBUTE: Represents properties of an entity
Example: user_id, report_id, timestamp

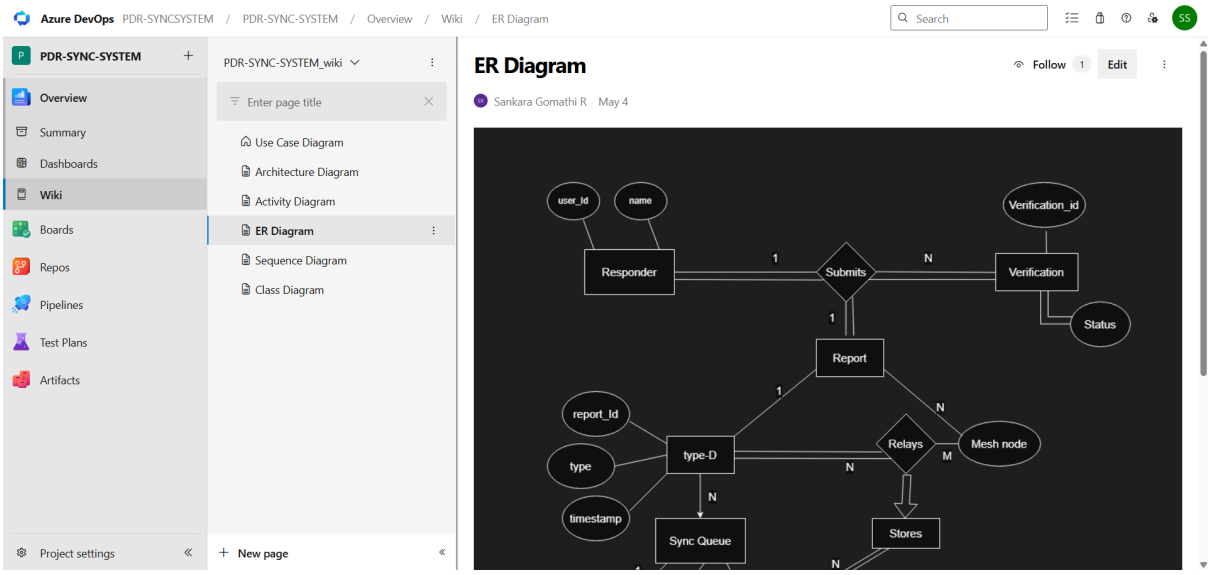
RELATIONSHIP: Represents interaction between entities
Example: Submits, Relays, Stores

CARDINALITY: Defines the number of relationships
Example: One-to-Many (1:N), Many-to-Many (M:N)

NOTATIONS USED:

NOTATION	DESCRIPTION

Rectangle	Entity
Oval	Attribute
Diamond	Relationship
Line	Connection
1, N, M	Cardinality



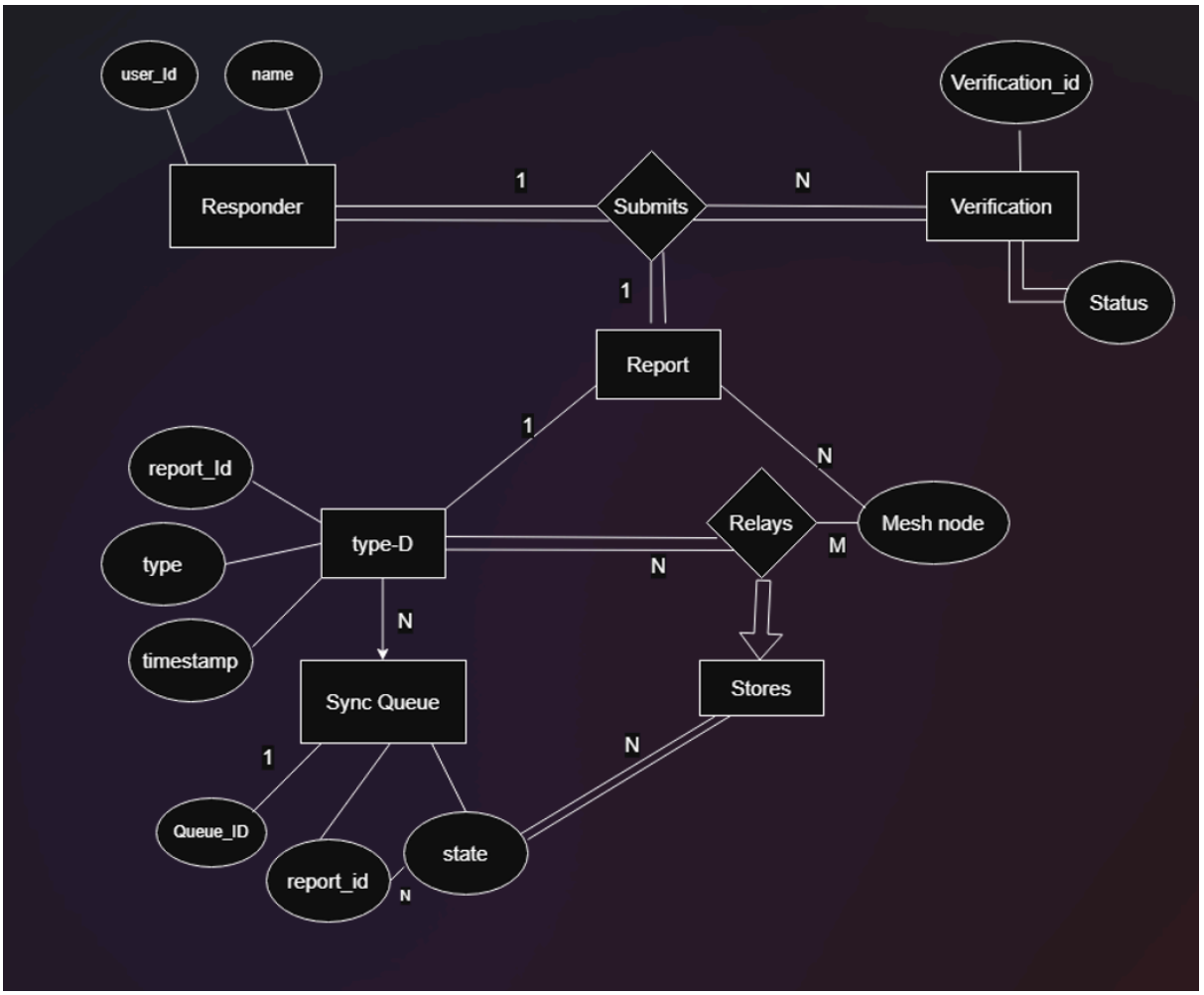
PROCEDURE:

1. Identify all entities in the system
   (Responder, Report, Verification, Mesh Node, Sync Queue)
2. Identify attributes for each entity
3. Define relationships between entities
4. Assign cardinality (1:N, M:N)
5. Draw entities using rectangles
6. Connect attributes using ovals
7. Represent relationships using diamonds
8. Validate diagram with system requirements

STEPS TO CREATE ER DIAGRAM IN AZURE:

1. Login to Azure Portal
2. Open Azure DevOps Project
3. Navigate to Repos or Wiki
4. Use diagram tools such as draw.io
5. Create entities and attributes

6. Define relationships and cardinalities
7. Export the diagram as an image
8. Upload to Azure DevOps



EXPLANATION OF ER DIAGRAM:

The ER diagram represents the data structure of the system.

RESPONDER

- Attributes: user_id, name
- A responder submits reports

REPORT

- Central entity representing disaster data
- Connected with multiple system components

VERIFICATION

- Attributes: verification_id, status
- Stores validation details of reports

TYPE-D (RESOURCE TYPE DATA)

- Attributes: report_id, type, timestamp
- Defines type of resource reported

SYNC QUEUE

- Attributes: queue_id, report_id, state
- Stores reports before synchronization

MESH NODE

- Represents devices participating in data relay

RELATIONSHIPS:

1. RESPONDER — SUBMITS — REPORT
    - One responder can submit many reports (1:N)
2. REPORT — VERIFICATION
    - One report can have multiple verifications
3. REPORT — RELAYS — MESH NODE
    - Many-to-many relationship (M:N)
4. REPORT — STORES — SYNC QUEUE
    - Reports are stored before syncing
5. TYPE-D — REPORT
    - Defines report details

RESULT:

Thus, the ENTITY RELATIONSHIP DIAGRAM for the system “PRECISION DISASTER RECOVERY (PDR-SYNC)” was successfully designed, representing entities, attributes, and relationships.